MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

AMPLIFIER MODULE

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/3(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope

This specification sheet provides performance and physical characteristics for the amplifier used in active Interphone Control panel C-2646 as an Audio Amplifier.

1.2 Military Part Number

The military part number shall be M22352/1-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Item Definition. The Amplifier Module shall provide amplification of audio signals as specified herein when used in active Interphone Control panel C-2626/AIC-14.

3.1.1 Item Diagram. The Amplifier Module shall provide the functions shown schematically on Figure 1.

3.1.2 Interface Definition. The Amplifier Module input and output signals will normally display the characteristics presented in Table I:

TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
No. 1 (NO. 1)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
No. 2 (NO. 2)	Voice and sidetone range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Audio (AUDIO)	Voice and sidetone range audio (300 to 6000 Hz) signal ranging from 0 to 16.9 Vac.
Grounds:	
Power (PWR GND)	Signal and power return. See 3.4.4.2 of MIL-I-22352(AS).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 No. 1. (NO. 1). The Amplifier Module shall have a NO. 1 audio input circuit having the following performance characteristics:

3.2.1.1.1.1 Signal Frequency. The amplifier shall accept input drive signals which range from 300 to 6,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The amplifier shall accept input drive signals which range in amplitude from 0.01 to 0.10 Vac. The amplifier shall be capable of withstanding overvoltage inputs of up to 1.0 Vac without sustaining damage. The amplifier shall be capable of operating within the limits specified herein when the signal input contains a dc component of up to 2 volts.

3.2.1.1.1.3 Input Impedance. The amplifier shall have an input impedance of 1200 ohms ±10 percent.

3.2.1.1.1.4 Isolation. The isolation between signal inputs shall be not less than 20 dB.

3.2.1.1.2 No. 2. (NO. 2). The NO. 2 input characteristic shall be as described in Section 3.2.1.1.1 for INPUT NO. 1.

3.2.1.1.3 Power (PWR). The Amplifier Module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the amplifier shall not exceed 100 milliamperes.

3.2.1.2 Output Characteristics. The Amplifier Module shall have the following output characteristics:

3.2.1.2.1 Audio (AUDIO). The AUDIO output shall have the following characteristics:

3.2.1.2.1.1 Frequency, Amplitude and Gain. The AUDIO signal output amplitude shall exceed the input amplitude(0.06 Vac) at 1,000 Hz by 43 ±2dB. The AUDIO output signal amplitude shall be in accordance with the response curves of Figure 2 when terminated in a 600 ohm resistive load. The input/output characteristic (gain) of the amplifier shall be linear within ±2dB when the input level is varied between 0.01 and 0.10 volts rms.

3.2.1.2.1.2 Power. The amplifier shall be capable of producing an output of 300 milliwatts into a non-inductive load of 270 to 600 ohms at 1,000 Hz. The amplifier shall be capable of producing an output of 150 milliwatts into a non-inductive load of 601 and 1,000 ohms at 1000 Hz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.

3.2.1.2.1.3 Output Impedance. The AUDIO output shall have a nominal impedance of 600 ohms ±10 percent.

3.2.1.2.1.4 Distortion. The total harmonic distortion over the frequency range of 300 to 6,000 Hz and input amplitude range of 0.01 to 0.10V shall not exceed 5 percent under standard conditions as defined in MIL-I-22352(AS) Appendix A. The total harmonic distortion over the frequency range of 300 to 6,000 Hz shall not exceed 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.2.1.2.1.5 Phasing. The phase difference between the NO. 1 input and the AUDIO output shall be 180 ± 20 degrees. The phase difference between the NO. 2 input and the AUDIO output shall be 180 ± 20 degrees.

3.2.1.2.1.6 Amplifier Stability. The amplifier which produces the AUDIO output signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.1.7 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope as defined in MIL-E-5400 for Class 2 equipment with the input circuit shorted to system ground.

3.2.1.2.1.8 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when terminated into a 600 ohm load within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz irrespective of orientation.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The Amplifier Module shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 4.5 ounces.

3.3 Design and Construction. The design and construction of the Amplifier Module shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS. See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
Navy - AS

Project No. 5831-N042-1

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:03Z
Check the source to verify that this is the current version before use.

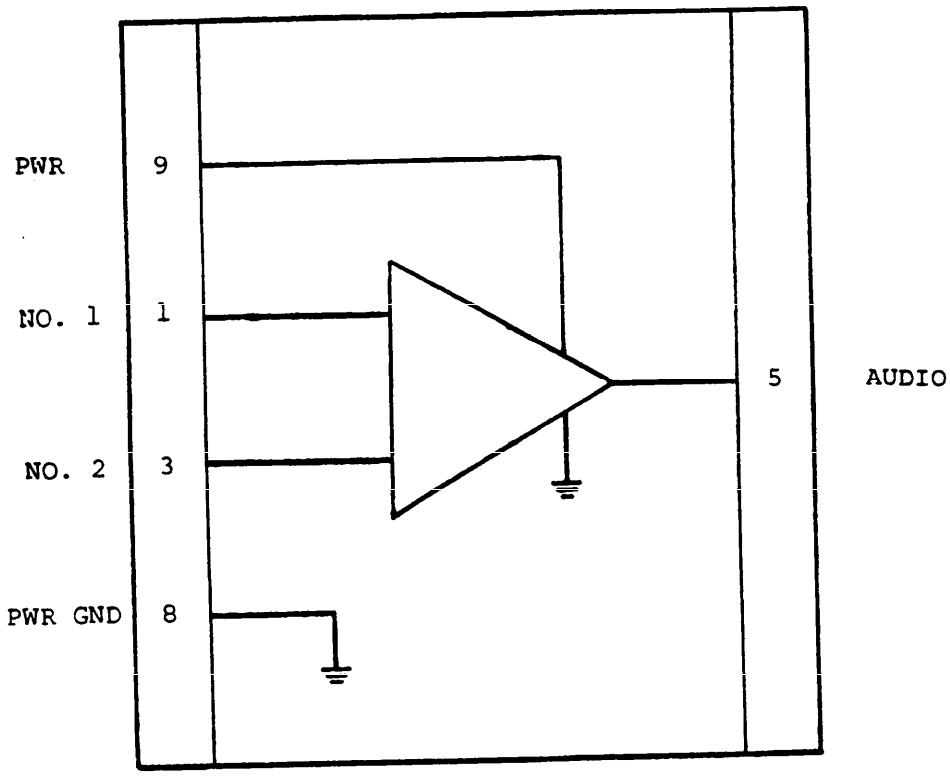
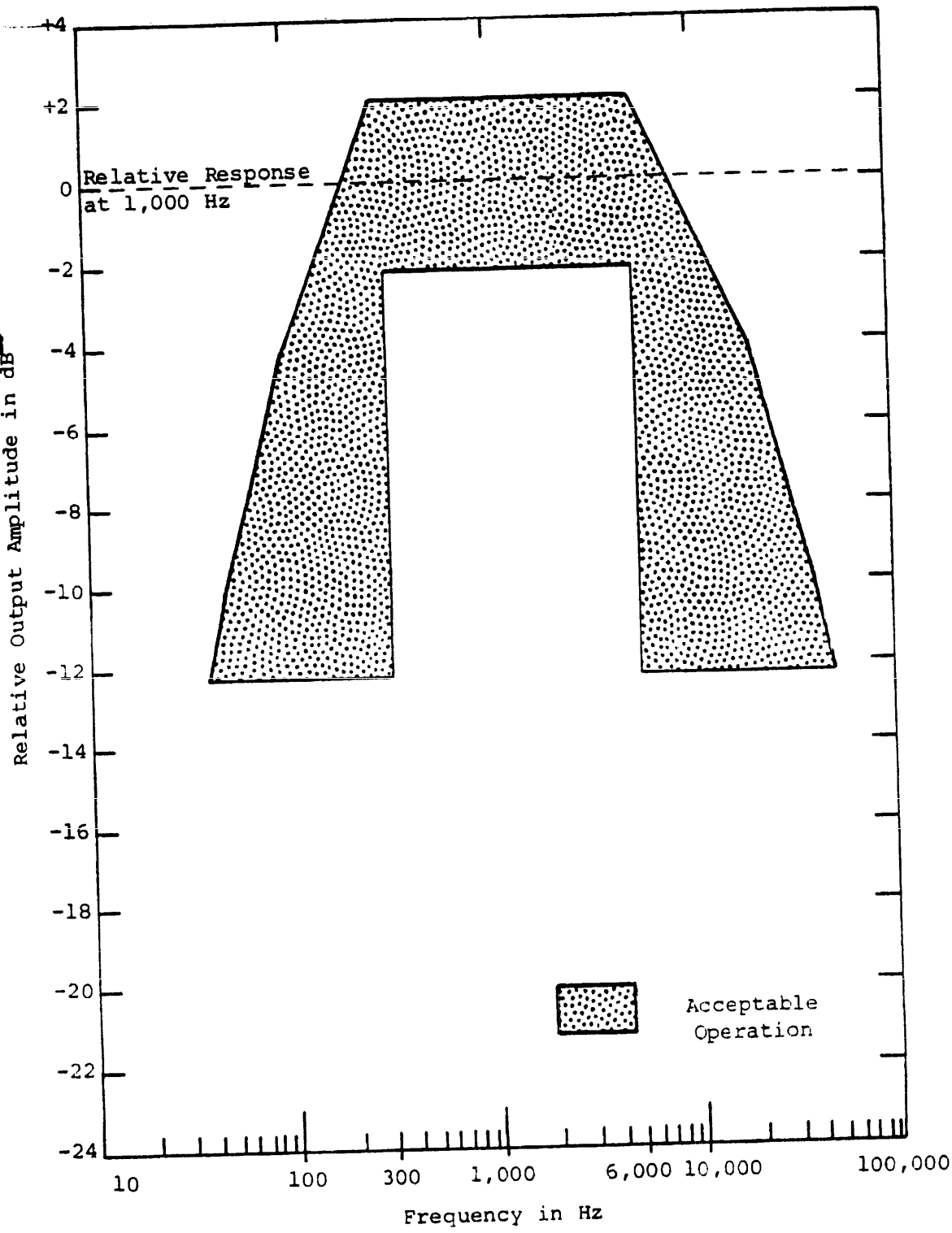


FIGURE 1. Amplifier module functional diagram.

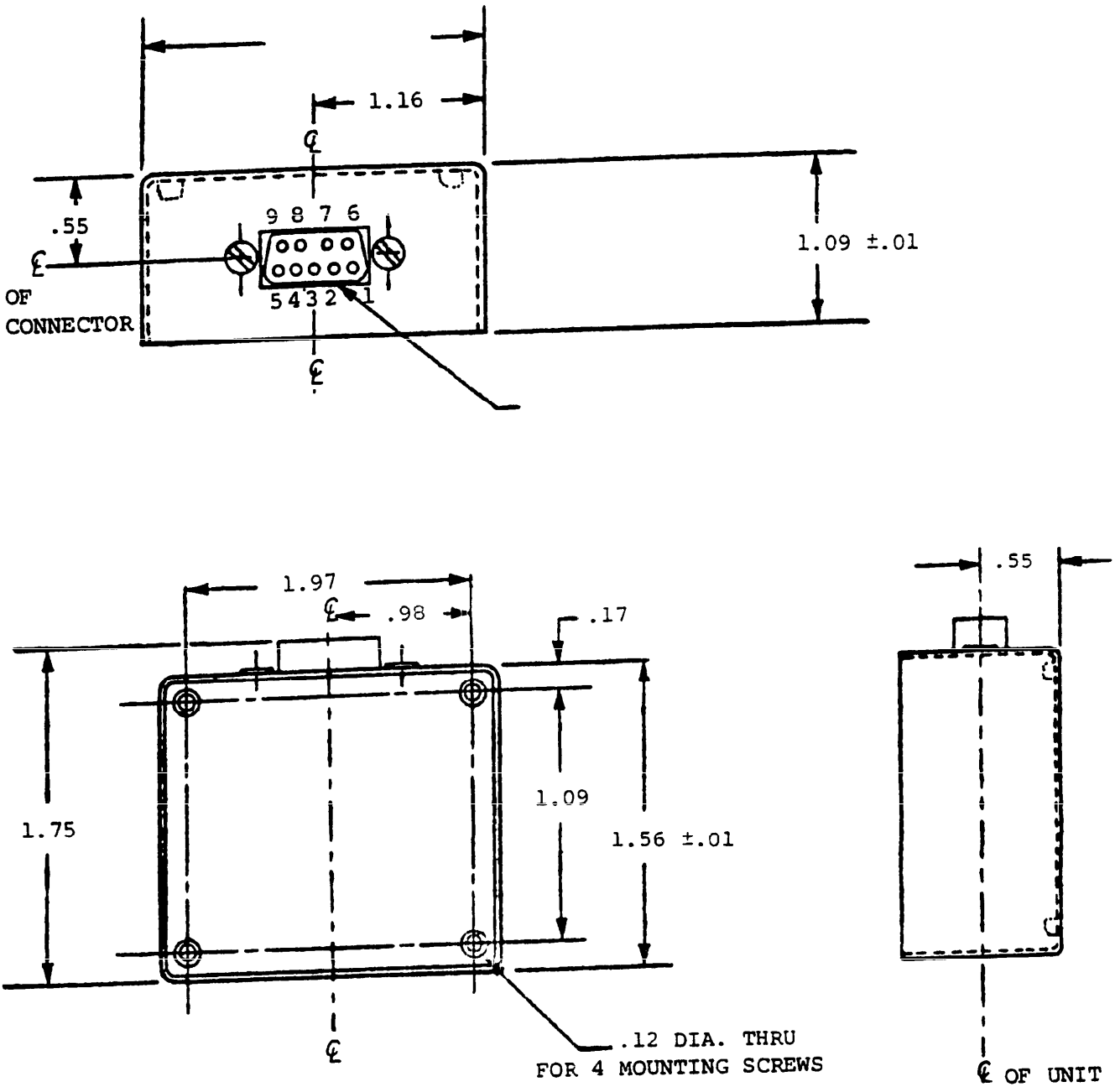
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.



1/ Termination = 600 ohms for AUDIO output.
2/ Output amplitude is measured with respect to the amplitude at 1,000 Hz.

FIGURE 2. Amplifier module frequency response.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:03Z
Check the source to verify that this is the current version before use.



1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 3. Amplifier module outline drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/1(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-AMPLIFIER MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Inclu Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil – Downloaded: 2026-Nov-19 12:03Z
Check the source to verify that this is the current version before use.

STRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in omitting comments and suggestions for improvements. All users of military standardization documents are invited to provide gestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and iled. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the oblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an knowledgement will be mailed to you within 30 days to let you know that your comments were received and are being sidered.

TE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of cification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

EPARTMENT OF THE NAVY
ommanding Officer
val Air Engineering Center
ngineering Specifications and Standards Department
(ESSD), Code 93
akehurst, NJ 08733

FICIAL BUSINESS
ALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil - Downloaded: 2026-10-19T12:03Z
Check the source to verify that this is the current version before use.